

DwarPal — Universal B2B SaaS Platform

A Multi-Vertical Gatepass, Visitor Access & Security Platform for Higher Education, K-12 Schools, Corporate Offices, and Manufacturing Industries.

Status: Approved Architecture Plan (v2.0) **Target Release:** Production Market Ready **Platform:** Node.js + React + MongoDB (Database-Per-Tenant)

1. Executive Summary & Value Proposition

DwarPal transforms institutional access operations from outdated paper slips and disconnected gate registers into a centralized, tamper-proof **B2B SaaS Platform**. Built with a multi-vertical dynamic workflow engine, DwarPal seamlessly adapts to the distinct governance, safety, and approval requirements of four major sectors:

Higher Education (Colleges & Universities)

Digital student out-passes, hostel night curfew enforcement, faculty leave substitution workflows, and multi-tier HOD/Principal/Warden approvals.

K-12 Schools

Verified parent student pickup verification, emergency early-exit approvals, class teacher consent, and automated parent exit SMS/email alerts.

Corporate Tech Parks & Offices

Employee working-hour check-outs, pre-scheduled VIP/client visitor passes, contractor badges, and seamless manager notification workflows.

Factories & Manufacturing Industries

Shift worker movement control, safety supervisor approvals, material/asset inward-outward returnable gatepasses, and contractor tracking.

2. Confirmed Strategic Architecture Matrix

PILLAR	SELECTED STRATEGY	ARCHITECTURAL EXECUTION & BENEFIT
Multi-Tenancy Model	Database-Per-Tenant	Dedicated MongoDB database (<code>dwarpal_org_[CODE]</code>) per organization within cluster. 100% hard data isolation and zero cross-tenant leak risk.
Free Trial & Anti-Spam	14-Day Trial + 6-Digit OTP	Prevents bot/spam databases. Verified admin email provisions dedicated DB. Auto-sleep on day 14 and automated 30-day purge of abandoned databases.
Member Access	Dual Login Routing	Universal code login at <code>app.dwarpal.com/login</code> + direct shareable links (<code>app.dwarpal.com/org/IITB</code>) for instant branded onboarding.
Workflow Engine	Step-by-Step Chain + Rules	Admin configures approval steps (Step 1 → Step 2 → Gate) with custom conditional triggers (e.g. Duration > 2 Days or Medical auto-approval).
Modular Passes	Toggleable Modules	Organizations turn on/off: Member Gatepasses, Visitor Access Passes, Staff Leave Swaps, and Inward/Outward Material/Goods Passes.
Security & Scanning	PWA Offline Camera Scanner	Security guard PWA scans HMAC-signed QR codes with local shift cache; syncs automatically even during intermittent gate Wi-Fi drops.

PILLAR	SELECTED STRATEGY	ARCHITECTURAL EXECUTION & BENEFIT
Commercial Model	Flexible SaaS Licensing	Accommodates upfront deployment fee + recurring maintenance/support, plus annual student/member batch renewal tiers.

3. Hybrid Database–Per–Tenant Architecture

DwarPal uses a two-tier database strategy that delivers both global platform scalability and uncompromising enterprise data privacy:

1. Global Platform Database (`dwarpal_global_core`)

- **Organization Registry:** Maps Org Codes (e.g. `APEX-2026`) to database names.
- **Subscriptions & Licences:** Status (`trial` , `active` , `expired`), expiry dates, and seat caps.
- **Global Email Index:** Fast lookup to identify which organization a logging-in user belongs to.
- **Platform SuperAdmin:** High-level tenant monitoring and billing management.

2. Dedicated Tenant Database (`dwarpal_org_[CODE]`)

- **Users & Profiles:** Student records, employee accounts, faculty credentials, guard profiles.
- **Gatepasses & Approvals:** Active out-passes, historical logs, approval remarks, checkout stamps.
- **Visitors & Materials:** Host invites, visitor kiosk logs, goods returnable records.
- **Departments & Shifts:** Custom departments, branches, wings, and approval chains.

4. Onboarding, Free Trial & Database Lifecycle

To eliminate wasted cluster storage from spam registrations while maximizing customer conversion, DwarPal implements a strict automated lifecycle:

- 1 **Organization Setup Wizard:** Admin enters Org Name, Vertical (College/School/Corporate/Factory), Official Email & Phone.
- 2 **Email OTP Gate:** System sends 6-digit verification code. Database is *NOT* created until OTP is successfully verified.
- 3 **Auto-Provisioning:** Global DB registers Org Code (e.g. `APEX-2026`) with `status: 'trial'` and `trialExpiresAt = now() + 14 days`. Dedicated DB (`dwarpal_org_APEX_2026`) is initialized and pre-seeded with vertical presets.
- 4 **Day 10 & 13 Automated Alerts:** Email reminders notify the Admin of trial expiration and provide direct upgrade links.
- 5 **Day 14 Expiration (Grace Period):** If not upgraded, tenant shifts to `trial_expired` (read-only for 15 days).
- 6 **Day 30 Automated Cleanup Cron:** Abandoned trial databases with zero subscription activity are permanently dropped, keeping the MongoDB cluster 100% clean.

5. Frictionless Member Access & Dual Routing

Direct Shareable Link (`/org/:code`)

Admins copy their dedicated link (e.g. `app.dwarpal.com/org/APEX-2026`) to share on notice boards, circulars, or WhatsApp. Opening this link pre-loads the organization's name, vertical badges, and login/register forms immediately.

Universal Org–Code Search

Users landing directly on `app.dwarpal.com/login` simply enter their Organization Code (`APEX-2026`). The browser remembers the last used Org Code in local cache for 1-click return visits.

6. Dynamic Approval Engine & Modular Pass Types

The Master Admin can customize approval hierarchies and toggle modules based on their institutional policies:

⚡ Step-by-Step Approval Chains

- **Step 1 (First Line):** Class Mentor, Shift Supervisor, or Team Manager review.
- **Step 2 (Escalation / High Level):** HOD, Hostel Warden, Plant Head, or Principal review.
- **Step 3 (Physical Verification):** Security Guard scan (OUT & IN timestamps).
- **Smart Rule Triggers:** Auto-escalate if duration > 2 days; auto-approve for verified medical emergencies.

📁 Modular Pass Modules (Toggleable)

- **Member Out-Passes:** Digital passes with rotating HMAC-signed tamper-proof QR codes.
- **Visitor & Guest Passes:** Host invitations, self-service kiosk check-ins, and AI screening.
- **Faculty / Staff Leaves:** Multi-day leave requests with peer workload substitution.
- **Material & Asset Passes:** Inward/outward returnable tools and equipment gate tracking.

7. Executive Master Admin Dashboard (Bird's Eye View)

🔒 100% Guaranteed Data Isolation

Each organization's Executive Dashboard queries ONLY its dedicated database. Principals, Directors, and HR Heads receive a real-time command center with absolute assurance of data privacy.

📊 Live Gate Counters

Real-time counts of members currently OUT, active visitors INSIDE campus, and overdue returnees.

📈 Peak Traffic Analytics

Hourly charts of gate check-outs/ins to optimize security staffing during peak departure windows.

👤 Bulk Directory Importer

Smart Excel/CSV import for 1,000+ students or employees with real-time format validation and error preview.

8. Phased Implementation Roadmap

PHASE	DELIVERABLE FOCUS	KEY MILESTONES
Phase 1	Core Multi-Tenancy & Global Platform	Global DB registry, Dynamic Mongoose connection pool, OTP-gated Org Setup Wizard, 14-day trial engine.
Phase 2	Dynamic Workflow & Master Admin	Step-by-step approval chain editor, modular pass toggles, Executive Bird's Eye View dashboard.
Phase 3	Member Onboarding & PWA Security	Dual routing (/org/ :code), Bulk Excel/CSV directory importer, PWA offline camera scanner.
Phase 4	Lifecycle Automation & Production Launch	Auto-purge cron worker for expired trial DBs, flexible licensing ledger, Dockerized CI/CD pipeline.

DwarPal SaaS Master Implementation Blueprint (v2.0) • Confidential • Designed for Scalable Institutional Access